

Disconnect Devices and Interlocks

Disconnect Devices

The facility is responsible for the robot's main disconnect device ensuring it complies with the correct electric codes.

Personnel servicing this equipment are responsible for the status of the robot's main disconnect device as specified on the facilities' lockout / tagout procedure.

Lockout/Tagout

The MagnaTran 7 utilizes an electrical power supply. Use of lockout / tagout procedures for the power supply when servicing the MagnaTran 7 is recommended by Brooks to ensure the safety of personnel servicing this robot.

Interlocks



WARNING

The MagnaTran 7 does not provide any personal safety or obstruction interlocks as a stand-alone unit.

However, safety interlocking capabilities exist for user safety. See [Safety Interlock on page 5-17](#) for instructions on connecting the robot motion emergency off safety interlock.

Operational interlocking capabilities exist through the discrete I/O. The flexibility of the interlocks is left up to the user to set up and manage. See [Operational Interlocks on page 6-23](#) in Chapter 6 for instructions on setting up the interlocks. Also see the Brooks Automation Marathon Express™ Cluster Tool Integration Platform User's Manual for additional interlocks if purchased with the Brooks System of components.

9 Maintenance and Repair

Overview

This chapter provides complete maintenance schedules and procedures for the Brooks Automation MagnaTran 7 Robot. The first section of this chapter provides preventive maintenance schedules and procedures. The second section of this chapter provides repair procedures for subsystem repair and replacement.

Brooks Automation offers training for troubleshooting and repair of the MagnaTran 7. Only qualified, properly trained persons should perform any maintenance or repair procedures.



PINCH POINT



HEAVY LIFTING



ELECTRICAL HAZARD

Crush points, pinch points, mechanical hazards, electrical hazards, shock hazards exist on the MagnaTran 7 robot. The procedures in this chapter should only be performed by qualified persons. Read and understand **Chapter 2: Safety before performing any procedure.**

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Wrist Band Adjustment

TOOLS:

- Small Phillips head screwdriver.
- Force gauge (.5 in / lbs. resolution)
- Dial Caliper or small 6" scale (1 / 16" resolution)

PROCEDURE:

1. Measure the wrist band thickness as shown in **Figure 9-4**. Use a small scale or dial caliper to determine the proper specification from Band Size Table.



CAUTION

While measuring the wrist bands, use caution so as not to nick or scratch the wrist bands.

Table 9-4: Band Size

Band Size	Specification
1 / 8"	1.5-3.0 in / lbs.
3 / 16"	3.0-4.5 in / lbs.
1 / 4"	3.5-5.0 in / lbs.

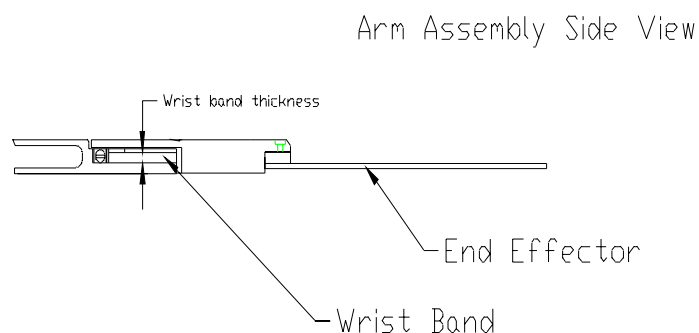


Figure 9-4: Arm Assembly Side View

2. Apply the plunger extension of the force gauge onto the tab located next to the spring loaded screw as shown in **Figure 9-5**. As soon as the tab moves slightly inward observe the force gauge reading.

If the force gauge reads less than specified in **Table 9-4**, adjust the spring loaded screw clockwise $1/8$ to $1/4$ turn at a time. Repeat step 2 of this procedure.

If the force gauge reads greater than specified amount adjust the spring loaded screw counter clock-wise $1/8$ to $1/4$ turns at a time. Repeat step 2 of this procedure.

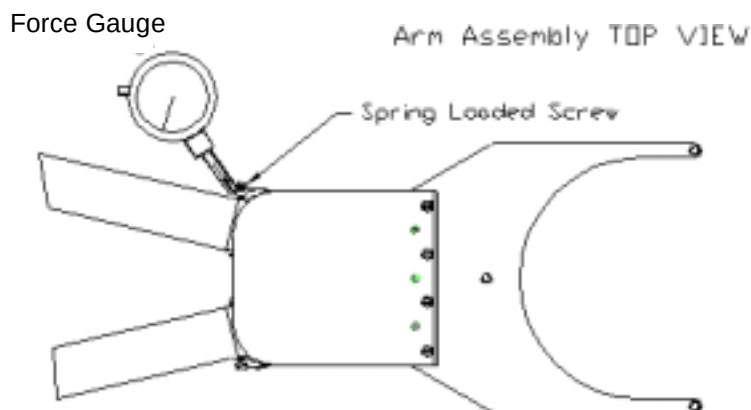


Figure 9-5: Arm Assembly Top View